

统计信号处理基础

第十二章

复合假设检验

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2026.5

内容概要

- 一、未知参数的影响
- 二、贝叶斯方法
- 三、广义似然比方法
- 四、应用案例
- 五、小结



一、未知参数的影响

WGN中未知信号检测

$$H_0 : x[n] = w[n]$$

$$H_1 : x[n] = A + w[n]$$

其中信号电平 A 是未知的。噪声 $w[n]$ 是方差为 σ^2 的WGN。如何检测是否存在信号？

采用NP准则，若似然比

$$L(\mathbf{x}) = \frac{p(\mathbf{x}; A, H_1)}{p(\mathbf{x}; H_0)} > \gamma$$

则判 H_1

$$L(\mathbf{x}) = \frac{p(\mathbf{x}; A, H_1)}{p(\mathbf{x}; H_0)} = \frac{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2\right\}}{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n]\right\}} > \gamma$$



$$-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 + \frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n] > \ln \gamma$$

$$-\frac{1}{2\sigma^2} \left\{ -2A \sum_{n=0}^{N-1} x[n] + NA^2 \right\} > \ln \gamma$$

$$A \sum_{n=0}^{N-1} x[n] > \sigma^2 \ln \gamma + \frac{NA^2}{2}$$

检测三要素：

- ✓ 检测统计量
- ✓ 判决方法
- ✓ 门限

1 若 $A > 0$

$$T(\mathbf{x}) = \frac{1}{N} \sum_{n=0}^{N-1} x[n] > \frac{\sigma^2}{NA} \ln \gamma + \frac{A}{2} = \gamma'$$



$$T(\mathbf{x}) = \frac{1}{N} \sum_{n=0}^{N-1} x[n] > \frac{\sigma^2}{NA} \ln \gamma + \frac{A}{2} = \gamma'$$

是否可实现？

$$T(\mathbf{x}) \sim \begin{cases} N\left(0, \frac{\sigma^2}{N}\right), & H_0 \\ N\left(A, \frac{\sigma^2}{N}\right), & H_1 \end{cases} \quad \longrightarrow \quad P_{FA} = \Pr(T(\mathbf{x}) > \gamma'; H_0) = Q\left(\frac{\gamma'}{\sqrt{\sigma^2/N}}\right)$$

与信号幅度无关——可实现！

$$\gamma' = \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA})$$

$$P_D = \Pr(T(\mathbf{x}) > \gamma'; H_1) = Q\left(\frac{\gamma' - A}{\sqrt{\sigma^2/N}}\right)$$

$$\longrightarrow P_D = Q\left(Q^{-1}(P_{FA}) - \sqrt{\frac{N}{\sigma^2}} A\right)$$

$$T(\mathbf{x}) = \frac{1}{N} \sum_{n=0}^{N-1} x[n] > \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA})$$

一致最大势检验
(uniformly most powerful test, UMP)

$$A \sum_{n=0}^{N-1} x[n] > \sigma^2 \ln \gamma + \frac{NA^2}{2}$$

2 若 $A < 0$

$$\sum_{n=0}^{N-1} x[n] < \frac{\sigma^2}{A} \ln \gamma + \frac{NA}{2} \quad \xrightarrow{T(\mathbf{x})} \quad \frac{1}{N} \sum_{n=0}^{N-1} x[n] < \frac{\sigma^2}{NA} \ln \gamma + \frac{A}{2} = \gamma'$$

$$T(\mathbf{x}) \sim N\left(0, \frac{\sigma^2}{N}\right), \quad H_0 \quad \xrightarrow{\quad} \quad P_{FA} = \Pr(T(\mathbf{x}) < \gamma'; H_0) = 1 - \Pr(T(\mathbf{x}) > \gamma'; H_0)$$

$$P_{FA} = 1 - Q\left(\frac{\gamma'}{\sqrt{\sigma^2/N}}\right) = Q\left(-\frac{\gamma'}{\sqrt{\sigma^2/N}}\right) \quad \xrightarrow{\quad} \quad \gamma' = -\sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA})$$

$$\text{即, 若 } T(\mathbf{x}) = \frac{1}{N} \sum_{n=0}^{N-1} x[n] < -\sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA}), \text{ 判 } H_1$$

单边检验

$$A > 0 \text{ 时, 若 } T(\mathbf{x}) = \frac{1}{N} \sum_{n=0}^{N-1} x[n] > \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA}), \text{ 判 } H_1$$

$-\infty < A < +\infty$?

双边检验

WGN中未知信号检测

$$H_0 : x[n] = w[n]$$

$$H_1 : x[n] = A + w[n]$$

其中信号电平 A 是未知的。噪声 $w[n]$ 是方差为 σ^2 的WGN。如何检测是否存在信号？

单边检验：

$$\begin{cases} H_0 : A = 0 \\ H_1 : A > 0 \end{cases} \text{ 或 } \begin{cases} H_0 : A = 0 \\ H_1 : A < 0 \end{cases} \quad \text{存在UMP}$$

双边检验：

$$\begin{cases} H_0 : A = 0 \\ H_1 : A \neq 0 \end{cases} \quad \text{不存在UMP}$$

推广至一般情况

单边检验：

$$\begin{cases} H_0 : \theta = \theta_0 \\ H_1 : \theta > \theta_0 \end{cases} \text{ 或 } \begin{cases} H_0 : \theta = \theta_0 \\ H_1 : \theta < \theta_0 \end{cases}$$

可能存在UMP

双边检验：

$$\begin{cases} H_0 : \theta = \theta_0 \\ H_1 : \theta \neq \theta_0 \end{cases}$$

不可能存在UMP

——求助于复合假设检验方法

复合假设检验的基本方法

贝叶斯方法

GLRT

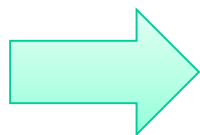


二、贝叶斯方法

未知参数:

$$H_0: \theta_0$$

$$H_1: \theta_1$$



数据PDF:

$$p(\mathbf{x}; H_0) = \int p(\mathbf{x} | \theta_0; H_0) p(\theta_0) d\theta_0$$

$$p(\mathbf{x}; H_1) = \int p(\mathbf{x} | \theta_1; H_1) p(\theta_1) d\theta_1$$

采用NP准则, 若似然比

$$L(\mathbf{x}) = \frac{p(\mathbf{x}; H_1)}{p(\mathbf{x}; H_0)} > \gamma$$

则判 H_1

方法特点:

- 该方法要求多重积分, 积分维数等于未知参数维数 —— 有时难求解
- 要求先验知识 —— 有时难把握
- 当具备条件时, 可以使用

例：WGN中未知信号检测——贝叶斯方法

$$H_0 : x[n] = w[n]$$

$$H_1 : x[n] = A + w[n]$$

其中信号电平 $A (-\infty < A < +\infty)$ 是未知的，其先验分布为 $N(0, \sigma_A^2)$ 。噪声 $w[n]$ 是方差为 σ^2 的WGN，且与信号统计独立。如何检测是否存在信号？

$$\begin{cases} H_0 : A = 0 \\ H_1 : A \neq 0 \end{cases} \quad \text{不存在UMP}$$

● 采用贝叶斯方法：

$$\left. \begin{aligned} p(\mathbf{x} | A; H_1) &= \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\} \\ p(A) &= \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp \left\{ -\frac{1}{2\sigma_A^2} A^2 \right\} \end{aligned} \right\} \rightarrow$$

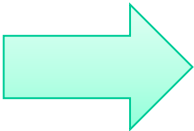
$$p(\mathbf{x}; H_1) = \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\} \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp \left\{ -\frac{1}{2\sigma_A^2} A^2 \right\} dA$$

$$p(\mathbf{x}; H_0) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n]\right\}$$

$$\begin{aligned} L(\mathbf{x}) &= \frac{p(\mathbf{x}; H_1)}{p(\mathbf{x}; H_0)} = \frac{\int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2\right\} \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left\{-\frac{1}{2\sigma_A^2} A^2\right\} dA}{\frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n]\right\}} \\ &= \frac{\int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2\right\} \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp\left\{-\frac{1}{2\sigma_A^2} A^2\right\} dA}{\exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n]\right\}} \\ &= \frac{1}{\sqrt{2\pi\sigma_A^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2} \left(\frac{N}{\sigma^2} + \frac{1}{\sigma_A^2} \right) A^2 + \frac{N}{\sigma^2} \bar{x} A\right\} dA \\ &= \frac{1}{\sqrt{2\pi\sigma_A^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{1}{2} \left(\frac{1}{\sigma_{A|x}^2} \left(A - \frac{N\bar{x}\sigma_{A|x}^2}{\sigma^2} \right)^2 - \frac{N^2\bar{x}^2\sigma_{A|x}^2}{\sigma^4} \right)\right\} dA \end{aligned}$$



$$\begin{aligned}
L(\mathbf{x}) &= \frac{p(\mathbf{x}; H_1)}{p(\mathbf{x}; H_0)} = \frac{1}{\sqrt{2\pi\sigma_A^2}} \int_{-\infty}^{\infty} \exp \left\{ -\frac{1}{2} \left(\frac{1}{\sigma_{A|x}^2} \left(A - \frac{N\bar{x}\sigma_{A|x}^2}{\sigma^2} \right)^2 - \frac{N^2\bar{x}^2\sigma_{A|x}^2}{\sigma^4} \right) \right\} dA \\
&= \frac{1}{\sqrt{2\pi\sigma_A^2}} \exp \left\{ \frac{N^2\bar{x}^2\sigma_{A|x}^2}{2\sigma^4} \right\} \int_{-\infty}^{\infty} \exp \left\{ -\frac{1}{2\sigma_{A|x}^2} \left(A - \frac{N\bar{x}\sigma_{A|x}^2}{\sigma^2} \right)^2 \right\} dA \\
&= \frac{\sqrt{2\pi\sigma_{A|x}^2}}{\sqrt{2\pi\sigma_A^2}} \exp \left\{ \frac{N^2\bar{x}^2\sigma_{A|x}^2}{2\sigma^4} \right\} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma_{A|x}^2}} \exp \left\{ -\frac{1}{2\sigma_{A|x}^2} \left(A - \frac{N\bar{x}\sigma_{A|x}^2}{\sigma^2} \right)^2 \right\} dA \\
&= \sqrt{\frac{\sigma_{A|x}^2}{\sigma_A^2}} \exp \left\{ \frac{N^2\bar{x}^2\sigma_{A|x}^2}{2\sigma^4} \right\}
\end{aligned}$$


 当 $\sqrt{\frac{\sigma_{A|x}^2}{\sigma_A^2}} \exp \left\{ \frac{N^2\bar{x}^2\sigma_{A|x}^2}{2\sigma^4} \right\} > \gamma$ 时, 判 H_1

仅保留与数据有关的项

$\bar{x}^2 > \gamma'$ 时, 判 H_1 , 即 $|\bar{x}| > \sqrt{\gamma'} = \gamma''$ 时, 判 H_1

——未知参数不再影响判决!



● 未知参数对检测性能的影响?

检测器: $|\bar{x}| > \sqrt{\gamma'} = \gamma''$

$$\bar{x} \sim \begin{cases} N\left(0, \frac{\sigma^2}{N}\right), H_0 \\ N\left(A, \frac{\sigma^2}{N}\right), H_1 \end{cases}$$

$$\begin{aligned} P_D &= \Pr(|\bar{x}| > \gamma''; H_1) \\ &= \Pr(\bar{x} > \gamma''; H_1) + \Pr(\bar{x} < -\gamma''; H_1) \\ &= Q\left(\frac{\gamma'' - A}{\sqrt{\sigma^2 / N}}\right) + 1 - Q\left(\frac{-\gamma'' - A}{\sqrt{\sigma^2 / N}}\right) \\ &= Q\left(\frac{\gamma'' - A}{\sqrt{\sigma^2 / N}}\right) + Q\left(\frac{\gamma'' + A}{\sqrt{\sigma^2 / N}}\right) \end{aligned}$$

$$\begin{aligned} P_{FA} &= \Pr(|\bar{x}| > \gamma''; H_0) \\ &= \Pr(\bar{x} > \gamma''; H_0) + \Pr(\bar{x} < -\gamma''; H_0) \\ &= 2Q\left(\frac{\gamma''}{\sqrt{\sigma^2 / N}}\right) \end{aligned}$$

$$\gamma'' = \sqrt{\frac{\sigma^2}{N}} Q^{-1}\left(\frac{P_{FA}}{2}\right)$$

$$P_D = Q\left(Q^{-1}\left(\frac{P_{FA}}{2}\right) - A\sqrt{\frac{N}{\sigma^2}}\right) + Q\left(Q^{-1}\left(\frac{P_{FA}}{2}\right) + A\sqrt{\frac{N}{\sigma^2}}\right)$$



- 假定未知参数已知时——透视检测器——其性能可作为上界

$A > 0$ 时

$$T(\mathbf{x}) = \bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] > \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA}) = \gamma'_+$$

$A < 0$ 时

$$T(\mathbf{x}) = \bar{x} = \frac{1}{N} \sum_{n=0}^{N-1} x[n] < -\sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA}) = \gamma'_-$$

✓ 当 $A > 0$ 时,

$$\bar{x} \sim \begin{cases} N\left(0, \frac{\sigma^2}{N}\right), H_0 \\ N\left(A, \frac{\sigma^2}{N}\right), H_1 \end{cases} \Rightarrow P_{FA} = \Pr(\bar{x} > \gamma'_+; H_0) = Q\left(\frac{\gamma'_+}{\sqrt{\sigma^2/N}}\right)$$

$$\gamma'_+ = \sqrt{\frac{\sigma^2}{N}} Q^{-1}(P_{FA})$$

$$P_D = \Pr(\bar{x} > \gamma'_+; H_1) = Q\left(\frac{\gamma'_+ - A}{\sqrt{\sigma^2/N}}\right) \rightarrow P_D = Q\left(Q^{-1}(P_{FA}) - A\sqrt{\frac{N}{\sigma^2}}\right)$$



✓ 同理，当 $A < 0$ 时， $P_D = Q\left(Q^{-1}(P_{FA}) + A\sqrt{\frac{N}{\sigma^2}}\right)$

✓ 当 $A > 0$ 时， $P_D = Q\left(Q^{-1}(P_{FA}) - A\sqrt{\frac{N}{\sigma^2}}\right)$

✓ 当 A 未知时， $P_D = Q\left(Q^{-1}\left(\frac{P_{FA}}{2}\right) - A\sqrt{\frac{N}{\sigma^2}}\right) + Q\left(Q^{-1}\left(\frac{P_{FA}}{2}\right) + A\sqrt{\frac{N}{\sigma^2}}\right)$

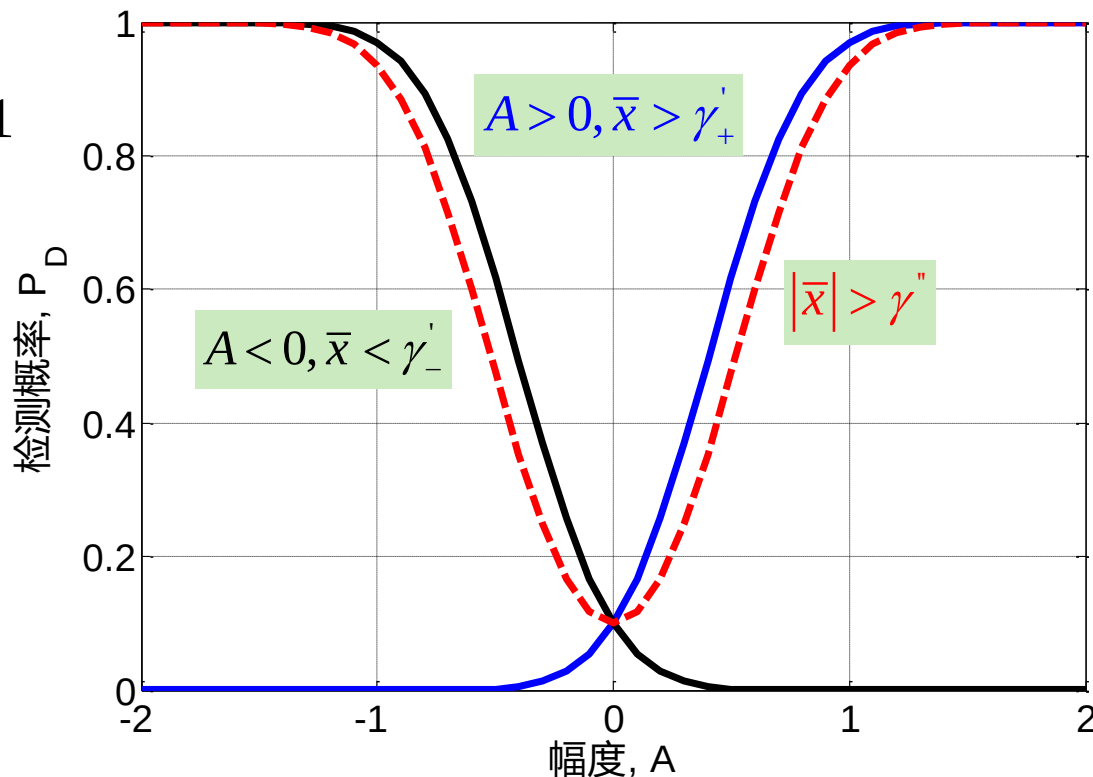
性能对比?

参数:

$N = 10, \sigma^2 = 1, P_{FA} = 0.1$

✓ 与透视检测器（假定未知参数已知时）相比，性能有所下降

✓ 次最佳检测器

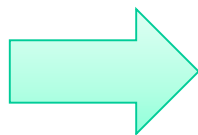


三、广义似然比方法

未知参数:

$$H_0: \theta_0$$

$$H_1: \theta_1$$



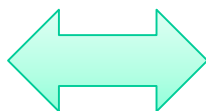
$$p(\mathbf{x}; \hat{\theta}_1, H_1)$$

$$p(\mathbf{x}; \hat{\theta}_0, H_0)$$

$$\text{MLE: } \hat{\theta}_0, \hat{\theta}_1$$

采用NP准则，若似然比

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{\theta}_1, H_1)}{p(\mathbf{x}; \hat{\theta}_0, H_0)} > \gamma$$



$$L_G(\mathbf{x}) = \frac{\max_{\theta_1} p(\mathbf{x}; \theta_1, H_1)}{\max_{\theta_0} p(\mathbf{x}; \theta_0, H_0)} > \gamma$$

则判 H_1

**Generalized Likelihood
Ratio Test (GLRT)**

- 核心思想: **最大**概率之比
- Vs 贝叶斯方法: **平均**概率之比

例：WGN中未知信号检测——GLRT

$$H_0 : x[n] = w[n]$$

$$H_1 : x[n] = A + w[n]$$

其中信号电平 $A(-\infty < A < +\infty)$ 是未知的。噪声 $w[n]$ 是方差为 σ^2 的WGN。如何检测是否存在信号？

$$\begin{cases} H_0 : A = 0 \\ H_1 : A \neq 0 \end{cases}$$

采用NP准则，若似然比

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, H_1)}{p(\mathbf{x}; H_0)} > \gamma$$

则判 H_1



$$p(\mathbf{x}; A, H_1) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\}$$

$$\frac{\partial \ln p(\mathbf{x}; A, H_1)}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)$$

$$\frac{\partial \ln p(\mathbf{x}; A, H_1)}{\partial A} = 0 \quad \left\} \begin{array}{l} \text{ } \\ \text{ } \end{array} \right. \Rightarrow \hat{A} = \bar{x}$$

$$\Rightarrow p(\mathbf{x}; \hat{A}, H_1) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - \bar{x})^2 \right\}$$

$$p(\mathbf{x}; H_0) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n] \right\}$$

$$\Rightarrow L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, H_1)}{p(\mathbf{x}; H_0)} = \frac{\exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - \bar{x})^2 \right\}}{\exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n] \right\}}$$



$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, H_1)}{p(\mathbf{x}; H_0)} = \frac{\exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - \bar{x})^2\right\}}{\exp\left\{-\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n]\right\}}$$

$$\begin{aligned} \ln L_G(\mathbf{x}) &= -\frac{1}{2\sigma^2} \left(\sum_{n=0}^{N-1} x^2[n] - 2\bar{x} \sum_{n=0}^{N-1} x[n] + N\bar{x}^2 - \sum_{n=0}^{N-1} x^2[n] \right) \\ &= -\frac{1}{2\sigma^2} (-2N\bar{x}^2 + N\bar{x}^2) \\ &= \frac{N\bar{x}^2}{2\sigma^2} \end{aligned}$$

仅保留与数据有关的项

$\bar{x}^2 > \gamma'$ 时, 判 H_1

即 $|\bar{x}| > \sqrt{\gamma'} = \gamma''$ 时, 判 H_1

Vs 贝叶斯方法

- 无需先验知识
- 运算简单
- 因此应用更广



例：WGN中未知信号检测——GLRT

$$H_0 : x[n] = w[n]$$

$$H_1 : x[n] = A + w[n]$$

其中信号电平 $A (-\infty < A < +\infty)$ 是未知的。噪声 $w[n]$ 是方差未知的WGN。如何检测是否存在信号？

$$\begin{cases} H_0 : A = 0, \sigma^2 \\ H_1 : A \neq 0, \sigma^2 \end{cases}$$

多余参数

采用NP准则，若广义似然比

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, \hat{\sigma}_1^2, H_1)}{p(\mathbf{x}; \hat{\sigma}_0^2, H_0)} > \gamma$$

则判 H_1

➤ H_1 时

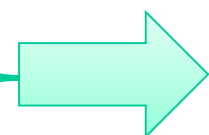
$$p(\mathbf{x}; A, \sigma^2, H_1) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)^2 \right\}$$

$$\frac{\partial \ln p(\mathbf{x}; A, \sigma^2, H_1)}{\partial A} = \frac{1}{\sigma^2} \sum_{n=0}^{N-1} (x[n] - A)$$

$$\frac{\partial \ln p(\mathbf{x}; A, \sigma^2, H_1)}{\partial A} = 0$$

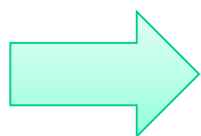
$$\frac{\partial \ln p(\mathbf{x}; A, \sigma^2, H_1)}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{n=0}^{N-1} (x[n] - A)^2$$

$$\frac{\partial \ln p(\mathbf{x}; A, \sigma^2, H_1)}{\partial \sigma^2} = 0$$



$$\hat{A} = \bar{x}$$

$$\hat{\sigma}_1^2 = \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - \bar{x})^2$$



$$p(\mathbf{x}; \hat{A}, \hat{\sigma}_1^2, H_1) = \frac{1}{(2\pi\hat{\sigma}_1^2)^{\frac{N}{2}}} \exp \left\{ -\frac{N}{2} \right\}$$

➤ H_0 时

$$p(\mathbf{x}; \sigma^2, H_0) = \frac{1}{(2\pi\sigma^2)^{\frac{N}{2}}} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=0}^{N-1} x^2[n] \right\}$$

$$\left. \begin{aligned} \frac{\partial \ln p(\mathbf{x}; \sigma^2, H_0)}{\partial \sigma^2} &= -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{n=0}^{N-1} x^2[n] \\ \frac{\partial \ln p(\mathbf{x}; \sigma^2, H_0)}{\partial \sigma^2} &= 0 \end{aligned} \right\} \Rightarrow \hat{\sigma}_0^2 = \frac{1}{N} \sum_{n=0}^{N-1} x^2[n]$$

$$\Rightarrow p(\mathbf{x}; \hat{\sigma}_0^2, H_0) = \frac{1}{(2\pi\hat{\sigma}_0^2)^{\frac{N}{2}}} \exp \left\{ -\frac{N}{2} \right\}$$

$$p(\mathbf{x}; \hat{A}, \hat{\sigma}_1^2, H_1) = \frac{1}{(2\pi\hat{\sigma}_1^2)^{\frac{N}{2}}} \exp \left\{ -\frac{N}{2} \right\} \Rightarrow$$

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{A}, \hat{\sigma}_1^2, H_1)}{p(\mathbf{x}; \hat{\sigma}_0^2, H_0)} = \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \right)^{\frac{N}{2}}, \quad \text{即} \quad 2 \ln L_G(\mathbf{x}) = N \ln \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \right)$$

$$2 \ln L_G(\mathbf{x}) = N \ln \left(\frac{\hat{\sigma}_0^2}{\hat{\sigma}_1^2} \right)$$

$$\left. \begin{aligned} \hat{\sigma}_0^2 &= \frac{1}{N} \sum_{n=0}^{N-1} x^2[n] \\ \hat{\sigma}_1^2 &= \frac{1}{N} \sum_{n=0}^{N-1} (x[n] - \bar{x})^2 \end{aligned} \right\} \Rightarrow \hat{\sigma}_1^2 = \frac{1}{N} \left(\sum_{n=0}^{N-1} x^2[n] + N\bar{x}^2 - 2 \sum_{n=0}^{N-1} x[n]\bar{x} \right)$$

$$= \hat{\sigma}_0^2 - \bar{x}^2 \Rightarrow \hat{\sigma}_0^2 = \hat{\sigma}_1^2 + \bar{x}^2$$

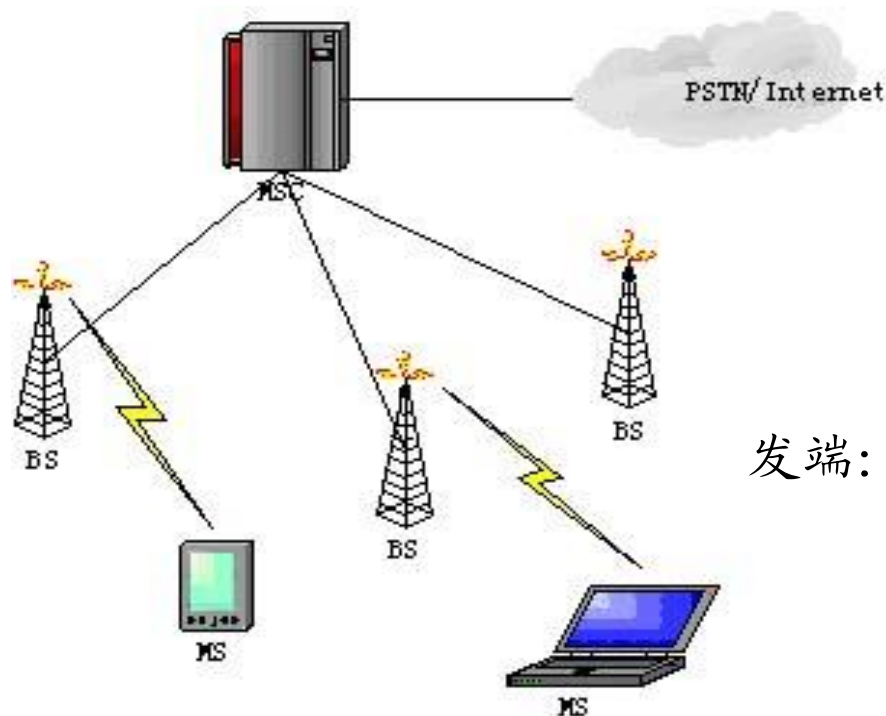
$$\Rightarrow 2 \ln L_G(\mathbf{x}) = N \ln \left(1 + \frac{\bar{x}^2}{\hat{\sigma}_1^2} \right) \Rightarrow T'(\mathbf{x}) = \frac{\bar{x}^2}{\hat{\sigma}_1^2} > \gamma'$$

门限:

$$\left. \begin{aligned} w[n] &\sim N(0, \sigma^2) \\ u[n] &\sim N(0, 1) \\ w[n] &= \sigma u[n] \end{aligned} \right\} \Rightarrow T'(\mathbf{x}; H_0) = \frac{\left(\frac{1}{N} \sum_{n=0}^{N-1} w[n] \right)^2}{\frac{1}{N} \sum_{n=0}^{N-1} (w[n] - \bar{w})^2} = \frac{\left(\frac{1}{N} \sigma \sum_{n=0}^{N-1} u[n] \right)^2}{\frac{1}{N} \sum_{n=0}^{N-1} (\sigma u[n] - \sigma \bar{u})^2} = \frac{\left(\frac{1}{N} \sum_{n=0}^{N-1} u[n] \right)^2}{\frac{1}{N} \sum_{n=0}^{N-1} (u[n] - \bar{u})^2}$$

四、应用案例

● DSSS系统信号检测问题



发端: $s(t) = PD(t)p(t)\cos(2\pi f_c t + \phi)$

收端: $r(t) = P'D(t-\tau)p(t-\tau)\cos(2\pi(f_c + f_D)(t-\tau) + \phi)$

收端: $r(t) = P' D(t - \tau) p(t - \tau) \cos(2\pi(f_c + f_D)(\underline{t - \tau}) + \phi)$

→ $r(t) = P' \underline{D(t - \tau)} p(t - \tau) \cos(2\pi(f_c + f_D)t + \phi')$

→ $r(t) = P'' p(t - \tau) \cos(2\pi(f_c + f_D)t + \phi')$

→ $r[n] = P'' p[n - n_0] \cos(2\pi(f_c + f_D)n + \phi')$

→ $x[n] = r[n] + w[n]$
 $= P'' p[n - n_0] \cos(2\pi(f_c + f_D)n + \phi') + w[n]$

重记为:

$x[n] = \textcolor{red}{A} p[n - \textcolor{red}{n}_0] \cos(2\pi(f_c + \textcolor{red}{f}_D)n + \textcolor{red}{\phi}) + w[n]$



两类假设:

$$H_0 : x[n] = w[n], \quad n = n_0, n_0 + 1, \dots, n_0 + M - 1$$

$$H_1 : x[n] = Ap[n - n_0] \cos(2\pi(f_c + f_D)n + \phi) + w[n], \quad n = n_0, n_0 + 1, \dots, n_0 + M - 1$$

其中信号电平 A 、相位 ϕ 、频率 f_D 、到达时间 n_0 是未知的。噪声 $w[n]$ 是方差为 σ^2 的WGN。如何检测是否存在信号?

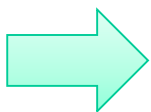
采用NP准则, 若广义似然比

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_1, H_1)}{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_0, H_0)} > \gamma$$

则GLRT判 H_1



$$p(\mathbf{x}; [A, n_0, f_D, \phi]^T, H_1) = \frac{1}{(2\pi\sigma^2)^{N/2}} \times \exp \left\{ -\frac{1}{2\sigma^2} \sum_{n=n_0}^{n_0+M-1} \left(x[n_0] - Ap[n-n_0] \cos(2\pi(f_c + f_D)n + \phi) \right)^2 \right\}$$



min: $J([A, n_0, f_D, \phi]^T) = \sum_{n=n_0}^{n_0+M-1} \left(x[n_0] - Ap[n-n_0] \cos(2\pi(f_c + f_D)n + \phi) \right)^2$

$$= \sum_{n=n_0}^{n_0+M-1} \begin{pmatrix} x[n_0] - \underline{A \cos(\phi)} p[n-n_0] \cos(2\pi(f_c + f_D)n) \\ \underline{-(-A \sin(\phi))} p[\underbrace{n-n_0}_{\alpha_2}] \sin(2\pi(f_c + f_D)n) \end{pmatrix}^2$$

$$= \sum_{n=n_0}^{n_0+M-1} \left(x[n_0] - \alpha_1 p[n-n_0] \cos(2\pi f_D n) - \alpha_2 p[n-n_0] \sin(2\pi f_D n) \right)^2$$



$$J([\alpha_1, \alpha_2, n_0, f_D]^T) = \sum_{n=n_0}^{n_0+M-1} \left(x[n] - \alpha_1 p[n-n_0] \cos(2\pi(f_c + f_D)n) - \alpha_2 p[n-n_0] \sin(2\pi(f_c + f_D)n) \right)^2$$

$$\mathbf{H} = \begin{bmatrix} p[0] \cos(2\pi(f_c + f_D)n_0) & p[0] \sin(2\pi(f_c + f_D)n_0) \\ p[1] \cos(2\pi(f_c + f_D)(n_0 + 1)) & p[1] \sin(2\pi(f_c + f_D)(n_0 + 1)) \\ \vdots & \vdots \\ p[M-1] \cos(2\pi(f_c + f_D)(n_0 + M - 1)) & p[M-1] \sin(2\pi(f_c + f_D)(n_0 + M - 1)) \end{bmatrix}$$

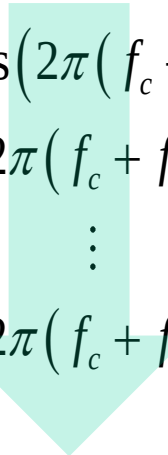
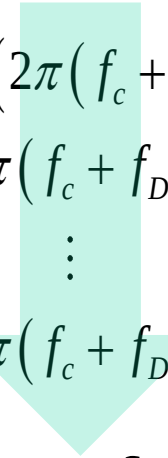
$$\boldsymbol{\theta} = \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix}$$

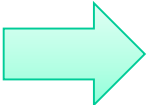
$$\begin{aligned} & \xrightarrow{\quad} J([\alpha_1, \alpha_2, n_0, f_D]^T) = (\mathbf{x} - \mathbf{H}\boldsymbol{\theta})^T (\mathbf{x} - \mathbf{H}\boldsymbol{\theta}) \\ & \xrightarrow{\quad} \hat{\boldsymbol{\theta}} = (\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x} \end{aligned}$$

$$J\left([\alpha_1, \alpha_2, n_0, f_D]^T\right) = (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}})^T (\mathbf{x} - \mathbf{H}\hat{\boldsymbol{\theta}}) \\ = \mathbf{x}^T \left(\mathbf{I} - \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \right) \mathbf{x}$$

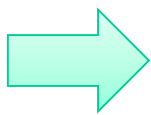
min: $J([\alpha_1, \alpha_2, n_0, f_D]^T)$  **max:** $J' = \mathbf{x}^T \mathbf{H}(\mathbf{H}^T \mathbf{H})^{-1} \mathbf{H}^T \mathbf{x}$

$$\mathbf{H} = \begin{bmatrix} p[0]\cos(2\pi(f_c + f_D)n_0) & p[0]\sin(2\pi(f_c + f_D)n_0) \\ p[1]\cos(2\pi(f_c + f_D)(n_0 + 1)) & p[1]\sin(2\pi(f_c + f_D)(n_0 + 1)) \\ \vdots & \vdots \\ p[M-1]\cos(2\pi(f_c + f_D)(n_0 + M - 1)) & p[M-1]\sin(2\pi(f_c + f_D)(n_0 + M - 1)) \end{bmatrix}$$

 $\mathbf{c}$  $\mathbf{s}$

 $J' = \begin{bmatrix} \mathbf{c}^T \mathbf{x} \\ \mathbf{s}^T \mathbf{x} \end{bmatrix}^T \begin{bmatrix} N/2 & 0 \\ 0 & N/2 \end{bmatrix}^{-1} \begin{bmatrix} \mathbf{c}^T \mathbf{x} \\ \mathbf{s}^T \mathbf{x} \end{bmatrix}$





$$J' = \frac{2}{N} \left\{ \left(\mathbf{c}^T \mathbf{x} \right)^2 + \left(\mathbf{s}^T \mathbf{x} \right)^2 \right\}$$

$$= \frac{2}{N} \left\{ \left(\sum_{n=0}^{M-1} \underset{\text{green}}{x[n_0 + n]} p[n] \cos \left(2\pi \left(\underset{\text{red}}{f_c} + \underset{\text{green}}{f_D} \right) (n_0 + n) \right) \right)^2 + \left(\sum_{n=0}^{M-1} \underset{\text{green}}{x[n_0 + n]} p[n] \sin \left(2\pi \left(\underset{\text{red}}{f_c} + \underset{\text{green}}{f_D} \right) (n_0 + n) \right) \right)^2 \right\}$$

采用NP准则，若广义似然比

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_1, H_1)}{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_0, H_0)} > \gamma$$

则GLRT判 H_1

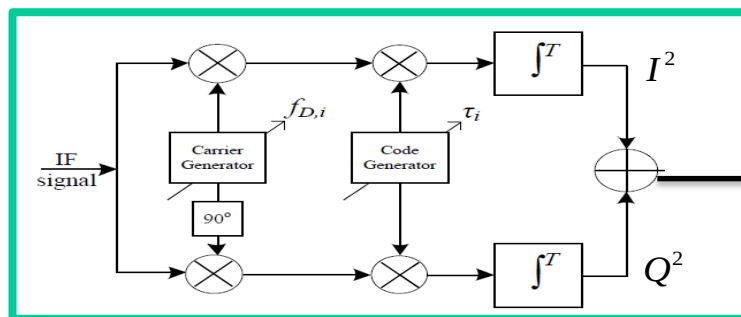
即，若

$$\max_{n_0, f_D} \left\{ \frac{2}{N} \left\{ \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \cos \left(2\pi (f_c + f_D)(n_0 + n) \right) \right)^2 + \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \sin \left(2\pi (f_c + f_D)(n_0 + n) \right) \right)^2 \right\} \right\} > \gamma' = \nu, \text{ 则判 } H_1$$

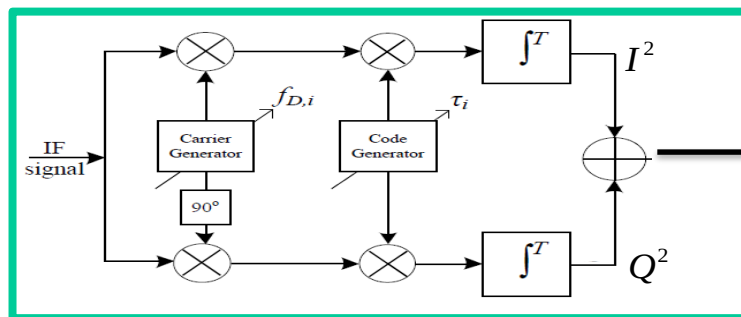


$$\max_{n_0, f_D} \left\{ \frac{2}{N} \left\{ \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \cos(2\pi(f_c + f_D)(n_0 + n)) \right)^2 + \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \sin(2\pi(f_c + f_D)(n_0 + n)) \right)^2 \right\} \right\} > \gamma' = v$$

频率格1



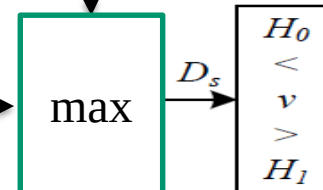
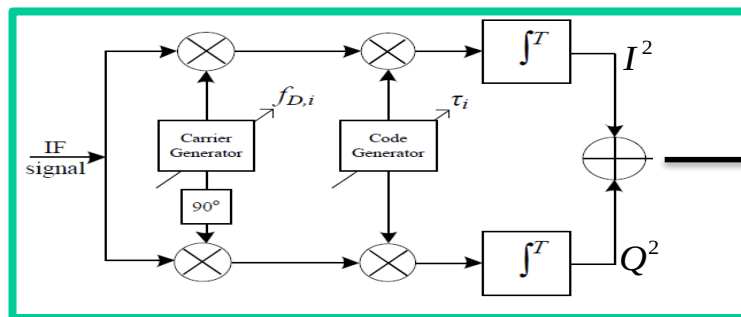
频率格2



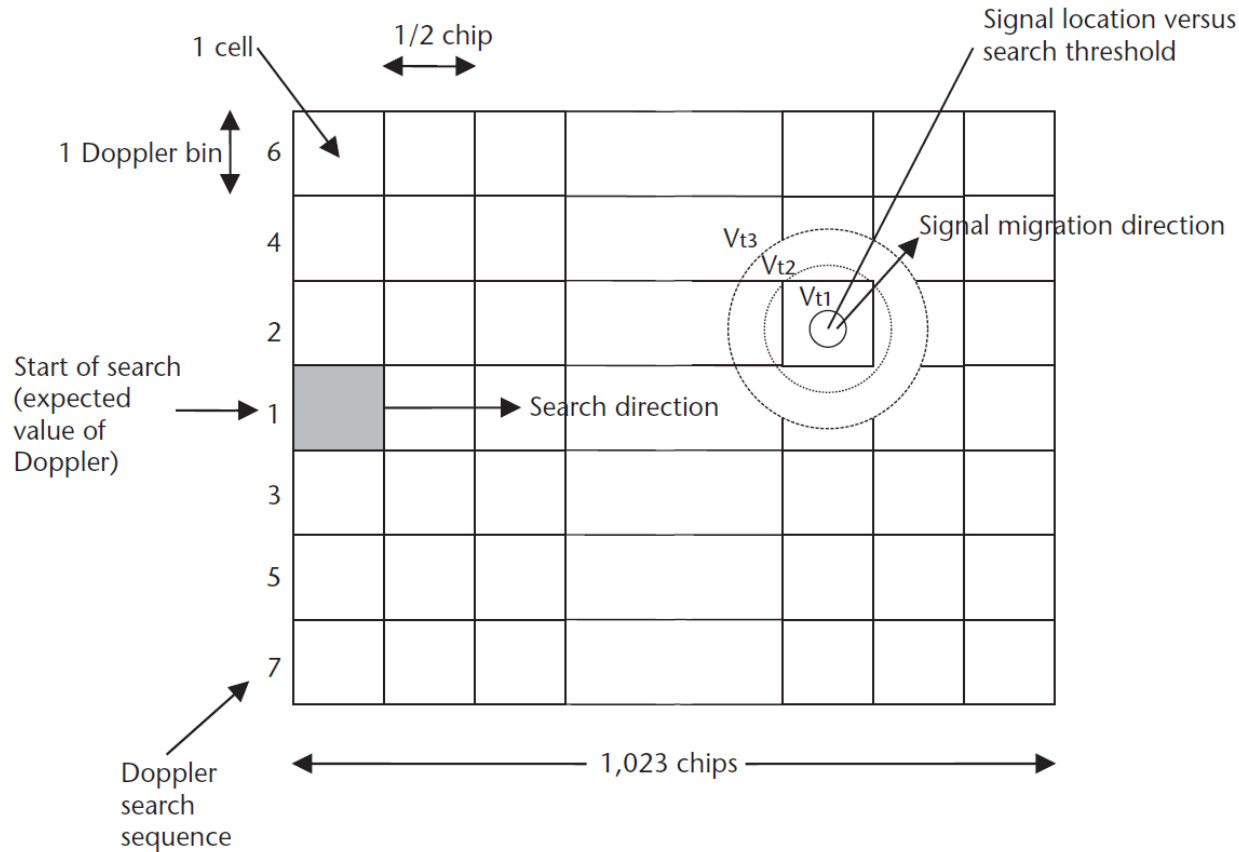
...

...

频率格K



$$\max_{n_0, f_D} \left\{ \frac{2}{N} \left\{ \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \cos(2\pi (f_c + f_D)(n_0 + n)) \right)^2 + \left(\sum_{n=0}^{M-1} x[n_0 + n] p[n] \sin(2\pi (f_c + f_D)(n_0 + n)) \right)^2 \right\} \right\} > \gamma' = v$$



图引用自 *Understanding GPS principles and application, Kaplan etc., figure 5.34, p220*

五、小结

■ 未知信号的检测

- 信号中含未知参数

——可推广至噪声中含未知参数场景

——称为复合假设检验

——当不含未知参数时称为简单假设检验

■ 一致最大势检测 (UMP)

- 当未知参数不影响检测器形成及其最优性时

■ 贝叶斯方法

- 需先验知识
- 需多重积分

$$L(\mathbf{x}) = \frac{\int p(\mathbf{x} | \boldsymbol{\theta}_1; H_1) p(\boldsymbol{\theta}_1) d\boldsymbol{\theta}_1}{\int p(\mathbf{x} | \boldsymbol{\theta}_0; H_0) p(\boldsymbol{\theta}_0) d\boldsymbol{\theta}_0} > \gamma$$

——平均概率之比

■ GLRT方法

- 无需先验知识
- 运算相对简单
- 应用广泛

$$L_G(\mathbf{x}) = \frac{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_1, H_1)}{p(\mathbf{x}; \hat{\boldsymbol{\theta}}_0, H_0)} > \gamma \iff L_G(\mathbf{x}) = \frac{\max_{\boldsymbol{\theta}_1} p(\mathbf{x}; \boldsymbol{\theta}_1, H_1)}{\max_{\boldsymbol{\theta}_0} p(\mathbf{x}; \boldsymbol{\theta}_0, H_0)} > \gamma$$

——最大概率之比

——这些方法对未知信号、未知噪声均适用

